

Model (Baseline Hazard) Comparisons

Setup

```
library(tidyverse)
library(arrow)
library(here)
library(rstanarm)
library(splines)
library(tinyplot)
library(survival)
library(survminer)
library(broom)
```

Load data

```
data <- read_parquet(here("data", "survival_complete.parquet"))
tinytheme("bw")
```

Fit models

```
CHAINS <- 4
CORES <- 4
SEED <- 809812
ITER <- 2000

mod1 <- stan_surv(
  formula = Surv(survival_time, event_death) ~ tx + ecog_fstcnt + diagnosis +
  ns(albumin, df = 2) + ns(c_reactive_protein, df = 2),
  basehaz = "ms", basehaz_ops = list(df = 5),
  data = data,
  chains = CHAINS,
  cores = CORES,
  seed = SEED,
  iter = ITER
)

mod_log <- stan_surv(
  formula = Surv(survival_time, event_death) ~ tx + ecog_fstcnt + diagnosis +
  log(albumin) + log(c_reactive_protein),
  basehaz = "ms", basehaz_ops = list(df = 5),
  data = data,
  chains = CHAINS,
  cores = CORES,
  seed = SEED,
  iter = ITER
)
```

```
print(mod1, digits = 3)
```

```
stan_surv
baseline hazard: M-splines on hazard scale
formula:      Surv(survival_time, event_death) ~ tx + ecog_fstcnt + diagnosis +
              ns(albumin, df = 2) + ns(c_reactive_protein, df = 2)
observations:  264
events:        40 (15.2%)
right censored: 224 (84.8%)
delayed entry: no
-----

```

	Median	MAD	SD	exp(Median)
(Intercept)	-3.238	1.060		NA
tx	-0.354	0.325		0.702
ecog_fstcnt1	0.457	0.493		1.579
ecog_fstcnt2	1.693	0.566		5.436
ecog_fstcnt3	2.629	0.612		13.866
diagnosis2	0.568	0.652		1.764
diagnosis3	1.552	0.751		4.721
diagnosis4	-0.219	0.914		0.804
diagnosis5	1.649	0.668		5.204
ns(albumin, df = 2)1	-1.177	1.286		0.308
ns(albumin, df = 2)2	0.203	1.081		1.225
ns(c_reactive_protein, df = 2)1	2.755	1.044		15.720
ns(c_reactive_protein, df = 2)2	0.810	0.842		2.248
m-splines-coef1	0.017	0.018		NA
m-splines-coef2	0.136	0.107		NA
m-splines-coef3	0.447	0.178		NA
m-splines-coef4	0.189	0.156		NA
m-splines-coef5	0.171	0.098		NA

```
-----
* For help interpreting the printed output see ?print.stanreg
* For info on the priors used see ?prior_summary.stanreg
```

```
mod1_exp <- update(mod1, basehaz = "exp", basehaz_ops = NULL)
mod1_weibull <- update(mod1, basehaz = "weibull", basehaz_ops = NULL)
# mod1_gompertz <- update(mod1, basehaz = "gompertz", basehaz_ops = NULL)
mod1_bspline <- update(mod1, basehaz = "bs", basehaz_ops = NULL)
mod1_mspline2 <- update(mod1, basehaz = "ms", basehaz_ops = list(df = 10))
```

Plot baseline hazards

```
plotfun <- function(model, title) {
  plot(model, plotfun = "basehaz") + # plot baseline hazard
  coord_cartesian(ylim = c(0, 0.007)) + # adjust y-axis limits
  labs(title = title) + # add plot title
  theme(plot.title = element_text(hjust = 0.5)) # centre plot title
}

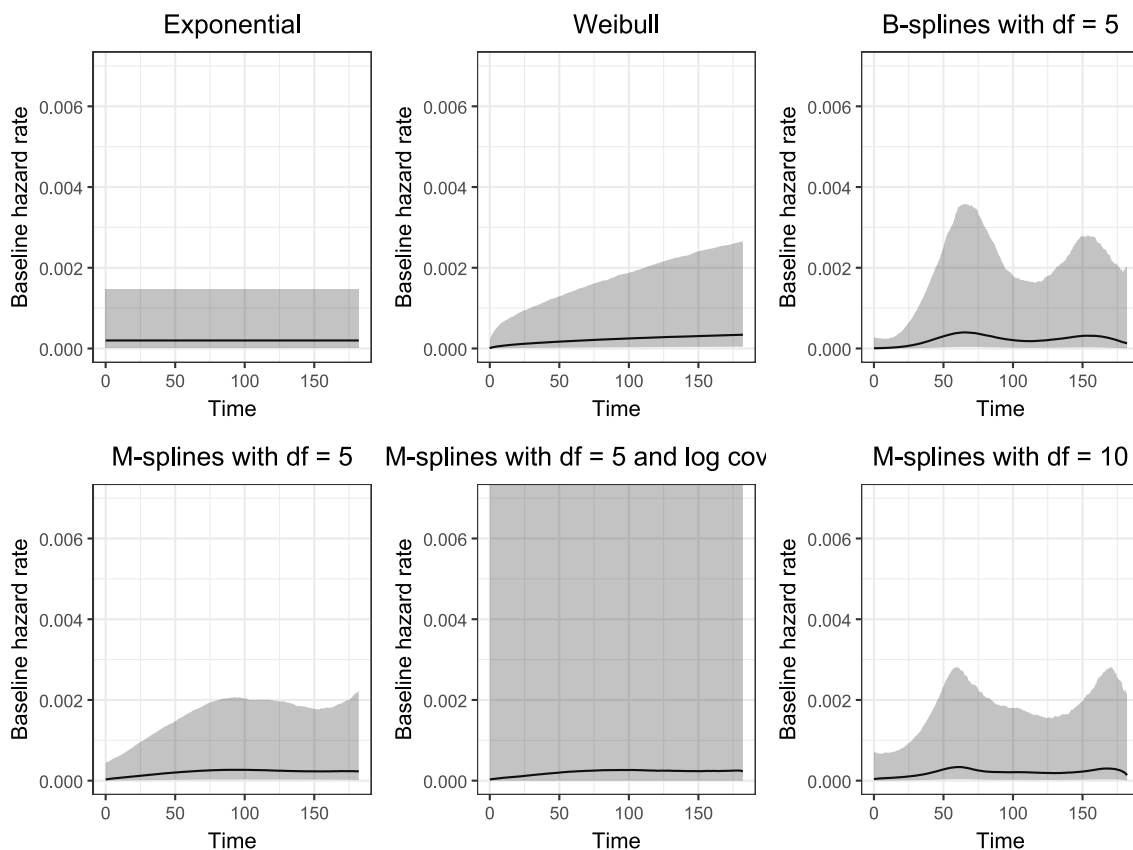
p_exp <- plotfun(mod1_exp, title = "Exponential")
```

```

p_weibull <- plotfun(mod1_weibull, title = "Weibull")
# p_gompertz <- plotfun(mod1_gompertz, title = "Gompertz")
p_bspline <- plotfun(mod1_bspline, title = "B-splines with df = 5")
p_mspline1 <- plotfun(mod1, title = "M-splines with df = 5")
p_mspline2 <- plotfun(mod1_mspline2, title = "M-splines with df = 10")
p_log <- plotfun(mod_log, title = "M-splines with df = 5 and log covariates")

bayesplot::bayesplot_grid(
  p_exp,
  p_weibull,
  # p_gompertz, # model didn't converge
  p_bspline,
  p_mspline1,
  p_log,
  p_mspline2,
  grid_args = list(ncol = 3)
)

```



Here's a simple explanation on the difference between flexible modelling of hazards or cumulative hazards and why the Royston-Parmar model doesn't work for Bayes:

<https://discourse.mc-stan.org/t/differences-between-splines-in-stan-surv-flexsurvspline/20787>

Regarding your first question. The RP models use cubic splines on the log cumulative hazard. The M-spline model in `stan_surv` uses cubic splines on the hazard. The B-spline model in `stan_surv` uses cubic splines on the log hazard. They all use slightly different splines, but that isn't the major difference, the major difference is what

scale they model things on (i.e. hazard vs log hazard vs log cumulative hazard). Since they are all trying to do something “flexible and wiggly” on the scale they choose, they are all trying to effectively achieve the same goal. And since the hazard, log hazard, and cumulative log hazard are all transformations of one another then doing something “flexible and wiggly” on one effectively translates to doing something “flexible and wiggly” on the other.

RP choose the log cumulative hazard because being of the cumulative hazard already means they don’t need to integrate the hazard to get the cumulative hazard (which is required for evaluating the likelihood) – meaning the RP approach is fast. But the log cumulative hazard has to increase – it is cumulative so it can’t go down! – but the splines RP use don’t enforce that, instead they let the data enforce it. This is ok for MLE, but for Bayesian sampling it is a problem, during MCMC the sampler might choose to make the log cumulative hazard go down, which is equivalent to a negative hazard and then the log hazard isn’t defined and the whole thing becomes a numerical disaster – the MCMC sampler will crash.

So we use M-splines on the hazard (M-splines are positive and always increase [i don't think M-splines always increase, just non-negative and integrates to 1]) or B-splines on the log hazard (can be negative, positive, go up or down). M-splines are easy to integrate to get the cumulative hazard, so they are faster. Whereas B-splines don’t integrate easily (we have to use numerical quadrature) so they are slower to fit the model. They are both “flexible and wiggly”, so in most situations (except for some edge cases) the M-splines will be flexible enough and therefore preferable to the faster model fitting.

So in short, the M-spline/B-spline choice is for numerical reasons, since RP models are problematic with MCMC. All are trying to do something “flexible and wiggly”, just on slightly different scales. So they should provide similar flexibility and results assuming you use a similar number of knots. There might be slight differences in the fit, and in some edge cases (e.g. a sharp change in the hazard early on or something) you might see one works better or worse, but with enough knots they should all do ok.

Compare models with LOO (Leave-One-Out Cross-Validation)

Not looking a gompertz as model didn’t converge

```
loo_exp <- loo(mod1_exp, k_threshold = 0.7)
loo_weibull <- loo(mod1_weibull, k_threshold = 0.7)
loo_bspline <- loo(mod1_bspline, k_threshold = 0.7)
loo_mspline1 <- loo(mod1, k_threshold = 0.7)
loo_mspline2 <- loo(mod1_mspline2, k_threshold = 0.7)
```

```
loo_compare(loo_exp, loo_weibull, loo_bspline, loo_mspline1, loo_mspline2)
```

	elpd_diff	se_diff
mod1_mspline2	0.0	0.0
mod1	-0.9	1.5
mod1_weibull	-1.1	2.1
mod1_bspline	-1.3	1.9
mod1_exp	-2.4	2.2

elpd_diff is not larger 4 and larger than 2x se_diff for any model, so no model is clearly better than the others based on LOO. Will therefore use M-splines with df = 5 as it is more flexible than exponential and weibull (i.e., will likely fit new data better) and more parsimonious than M-splines with df = 10.

We now refit the chosen model with the default priors hardcoded for improved reproducibility. We will also use a more conservative prior for the treatment effect (tx). We deem it unlikely that invitation to the single arm trial (SAT) could increase or decrease hazard ratio (HR) by more than a factor of 2, i.e., a HR of 0.5 or 2.0. To place ca. 80% of the prior probability mass within this range we use a normal prior on the log-hazard ratio scale with mean 0 and standard deviation 0.55.

```
tx_scale <- 0.55
default_scale <- 2.5
my_prior <- normal(
  location = c(rep(0, 12)),
  scale = c(tx_scale, rep(default_scale, 11))
)

mod2 <- stan_surv(
  formula = Surv(survival_time, event_death) ~ tx + ecog_fstcnt + diagnosis +
ns(albumin, df = 2) + ns(c_reactive_protein, df = 2),
  basehaz = "ms",
  basehaz_ops = list(df = 5),
  prior = my_prior,
  data = data,
  chains = CHAINS,
  cores = CORES,
  seed = SEED,
  iter = ITER
)
```

Calculate marginal treatment effects

Survial probability

```
# --- Marginal (Counterfactual) Survival Predictions ---

# We are going to use g-computation to estimate the marginal (population average)
# Rstanarm has built in standarization that averages across covariates for us
# (but I don't think you can specifiy which ones, so we do it once for everyone
# treated, once for everyone not treated and manually combine)

nd_0 <- data |>
  mutate(tx = 0) |>
  select(tx, ecog_fstcnt, diagnosis, albumin, c_reactive_protein)

ps_0 <- posterior_survfit(
  mod2,
  newdata = nd_0,
  times = 0,
  extrapolate = TRUE,
  standardise = TRUE,
  control = list(edist = 7000, epoints = 200),
  draws = 4000
```

```

)

nd_1 <- data |>
  select(ecog_fstcnt, diagnosis, albumin, c_reactive_protein) |>
  mutate(tx = 1)

ps_1 <- posterior_survfit(
  mod2,
  newdata = nd_1,
  times = 0,
  extrapolate = TRUE,
  standardise = TRUE,
  control = list(edist = 7000, epoints = 200),
  draws = 4000
)

untrt <- ps_0 |> data.frame() |> mutate(group = "Untreated")
trt <- ps_1 |> data.frame() |> mutate(group = "Treated")

marginal_data <- bind_rows(untrt, trt)
marginal_data$group <- factor(
  marginal_data$group,
  levels = c("Untreated", "Treated")
)

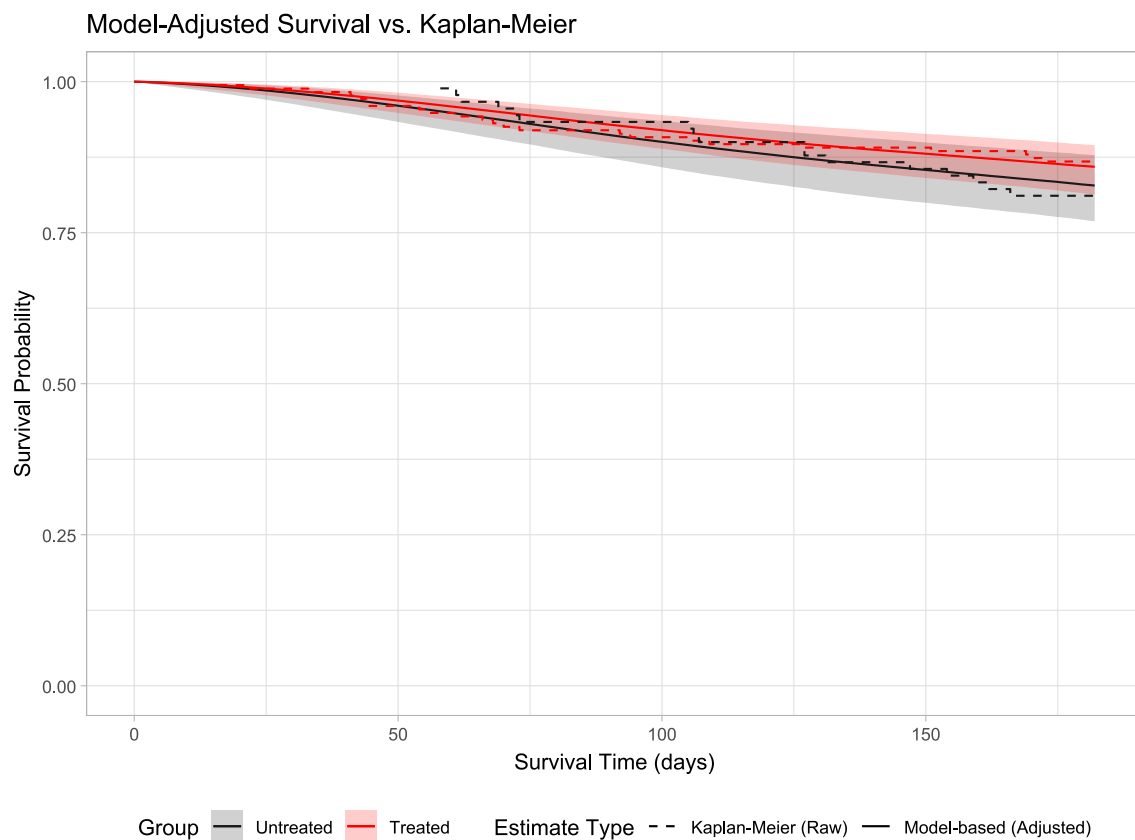
```

```

# --- Compare to Kaplan-Meier estimates ---
km_fit <- survfit(Surv(survival_time, event_death) ~ tx, data = data)
km_data <- tidy(km_fit) |>
  mutate(group = ifelse(strata == "tx=1", "Treated", "Untreated"))

ggplot(marginal_data, aes(x = time, y = median, group = group, color = group)) +
  # Model's 95% Credible Interval
  geom_ribbon(aes(ymin = ci_lb, ymax = ci_ub, fill = group), alpha = 0.2, color = NA) +
  # Model's median survival curve (solid line)
  geom_line(aes(linetype = "Model-based (Adjusted)")) +
  # Kaplan-Meier curve (dashed line)
  geom_step(data = km_data, aes(x = time, y = estimate, linetype = "Kaplan-Meier
(Raw)")) +
  # Aesthetics and labels
  scale_color_manual(values = c("Untreated" = "gray10", "Treated" = "red")) +
  scale_fill_manual(values = c("Untreated" = "gray10", "Treated" = "red")) +
  scale_linetype_manual(values = c("Model-based (Adjusted)" = "solid", "Kaplan-Meier
(Raw)" = "dashed")) +
  coord_cartesian(ylim = c(0, 1)) +
  labs(
    title = "Model-Adjusted Survival vs. Kaplan-Meier",
    x = "Survival Time (days)",
    y = "Survival Probability",
    color = "Group",
    fill = "Group",
    linetype = "Estimate Type"
  ) +
  theme_light() +
  theme(legend.position = "bottom")

```



The model based adjusted survival curves are very similar to the Kaplan-Meier estimates. Note that the KM-curves cross, but this is due to chance only, because we now that any difference in the data to plan our analyses is random.